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Finerenone in Persons with Chronic Kidney Disease without Diabetes

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ABSTRACT

BACKGROUND

In randomized trials, finerenone, a nonsteroidal mineralocorticoid receptor antagonist, improved kidney and cardiovascular outcomes in patients with type 2 diabetes and chronic kidney disease (CKD). Whether finerenone has similar effects in patients without diabetes who have CKD is unknown.

METHODS

We randomly assigned adults without diabetes who had CKD (estimated glomerular filtration rate [eGFR], 25 to <90 ml per minute per 1.73 m² of body-surface area) and albuminuria (urinary albumin-to-creatinine ratio, 200 to ≤3500, with albumin in milligrams and creatinine in grams) and were using a renin–angiotensin system inhibitor to receive finerenone (10 or 20 mg daily) or placebo. The primary outcome was the total eGFR slope (mean annual rate of change in the eGFR from baseline to month 32), assessed with a two-slope linear spline mixed-effects model. Secondary outcomes included a composite of kidney or cardiovascular events (reduction from baseline of at least 57% in the eGFR, kidney failure, hospitalization for heart failure, or death from cardiovascular causes), a composite of the two kidney events, and a composite of the two cardiovascular events.

RESULTS

A total of 1584 participants underwent randomization — 793 were assigned to the finerenone group, and 791 to the placebo group. The mean (±SD) baseline eGFR was 46.8±16.2 ml per minute per 1.73 m² with finerenone and 46.6±16.0 ml per minute per 1.73 m² with placebo. The mean annual rate of change in the eGFR from baseline to month 32 was −3.3 ml per minute per 1.73 m² (95% confidence interval [CI], −3.6 to −3.1) with finerenone and −4.0 ml per minute per 1.73 m² (95% CI, −4.3 to −3.8) with placebo (difference, 0.7; 95% CI, 0.3 to 1.1; P<0.001). Prespecified hierarchical testing showed that the risk of a composite kidney or cardiovascular outcome event was lower with finerenone than with placebo (hazard ratio, 0.77; 95% CI, 0.60 to 0.99; P=0.04); the hazard ratio was 0.78 (95% CI, 0.60 to 1.01) for the composite of the two kidney events and 0.60 (95% CI, 0.27 to 1.33) for the composite of the two cardiovascular events. The most common adverse event was hyperkalemia (135 participants [17.0%] with finerenone and 105 [13.3%] with placebo); hyperkalemia events led to discontinuation of the trial regimen in 12 participants (1.5%) and 1 participant (0.1%), respectively, and to hospitalization in 7 (0.9%) and 5 (0.6%).

CONCLUSIONS

Among adults with CKD who did not have diabetes, finerenone led to a slower decrease in the eGFR than placebo over 32 months. (Funded by Bayer; FIND-CKD ClinicalTrials.gov number, NCT05047263.)

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CME





THE PREVALENCE OF CHRONIC KIDNEY disease (CKD) has more than doubled since 1990, with approximately 800 million adults affected globally in 2023.¹ Although type 2 diabetes is the most common cause of CKD,² more than half the cases are in persons without diabetes.^{3,4} Renin–angiotensin system (RAS) inhibitors and sodium–glucose cotransporter 2 (SGLT2) inhibitors reduce the loss of kidney function and risk of kidney failure in adults with CKD, regardless of whether they have diabetes.^{5–9} Despite the use of these guideline-recommended treatments,¹⁰ progressive loss of kidney function and adverse cardiovascular outcomes occur in many persons with CKD, particularly those with severely increased albuminuria levels.¹¹ Thus, new therapies are needed.

Overactivation of the mineralocorticoid receptor is implicated in the pathophysiology of CKD through increased sodium and fluid retention and activation of proinflammatory and profibrotic processes.¹² Finerenone is a selective nonsteroidal mineralocorticoid receptor antagonist that has been shown to reduce the loss of kidney function and the risk of adverse kidney and cardiovascular outcomes in persons with CKD and type 2 diabetes.^{13–15} The current trial, FIND-CKD (Finerenone in Non-Diabetic CKD), examined the efficacy and safety of finerenone in adults without diabetes who had CKD.

METHODS

TRIAL DESIGN AND OVERSIGHT

We conducted a phase 3, international, prospective, double-blind, randomized trial of finerenone as compared with placebo in adults without diabetes who had CKD. The trial design was published previously¹⁶; the protocol is available with the full text of this article at NEJM.org. The trial was designed by the steering committee and Bayer (the sponsor). The trial followed the principles of the Declaration of Helsinki and the Council for International Organizations of Medical Sciences as well as the International Council for Harmonisation guidelines for Good Clinical Practice. An independent data and safety monitoring committee oversaw the trial throughout. Data were collected by the investigators, analyzed by the sponsor, and independently confirmed by staff at the University Medical Center Groningen using

the original data. The first author prepared the initial draft of the manuscript and had full access to the data. All the authors reviewed and revised subsequent drafts, approved the final version, and, together with the sponsor, agreed to submit the manuscript for publication. The first and last authors vouch for the accuracy and completeness of the data; the sponsor and the investigators vouch for the fidelity of the trial to the protocol.

PARTICIPANTS

Eligible participants had a clinical diagnosis of CKD and an estimated glomerular filtration rate (eGFR) of at least 25 but less than 60 ml per minute per 1.73 m² of body-surface area and a urinary albumin-to-creatinine ratio between 200 and 500 (with the albumin level measured in milligrams and the creatinine level measured in grams), or an eGFR of at least 25 but less than 90 ml per minute per 1.73 m² and a urinary albumin-to-creatinine ratio between 500 and 3500. Eligibility criteria included a serum potassium level of 4.8 mmol per liter or less and receipt of a stable dose of an angiotensin-converting enzyme inhibitor or an angiotensin-receptor blocker therapy for at least 4 weeks before screening. Patients were excluded if they had received a diagnosis of type 1 or type 2 diabetes or had a glycated hemoglobin level of at least 6.5%. Patients were also excluded if they had an indication for a steroidal mineralocorticoid receptor antagonist; had polycystic kidney disease, lupus nephritis, or antineutrophil cytoplasmic autoantibody–associated vasculitis; or had received immunosuppressive therapy within 6 months before screening. Full eligibility criteria are provided in Table S1 in the Supplementary Appendix, available at NEJM.org. All the participants provided written informed consent before trial entry.

TRIAL VISITS

Trial visits were prespecified to occur at screening, baseline, and months 1, 3, 6, 9, and 12. Thereafter, visits were prespecified to occur every 4 months until the end of the trial. Participants were to be treated for at least 32 months. An additional visit occurred 4 weeks after discontinuation of the trial regimen to assess the effects of finerenone as compared with placebo after treatment. At all trial visits, blood samples

were collected for eGFR and serum potassium assessments at the central laboratory; further details about the trial procedures are provided in the Supplementary Appendix.

TRIAL PROCEDURES

Participants were randomly assigned in a 1:1 ratio to receive oral finerenone or matching placebo. Randomization was conducted with the use of interactive response technology and included baseline SGLT2 inhibitor use (yes vs. no) and screening urinary albumin-to-creatinine ratio (≤ 1000 vs. >1000) as stratification factors. Finerenone therapy was initiated at a dose of 20 mg per day if the screening eGFR was at least 60 ml per minute per 1.73 m^2 and at 10 mg per day if the screening eGFR was 25 to less than 60 ml per minute per 1.73 m^2 . An increase in the dose of finerenone from 10 mg per day to the target dose of 20 mg per day was permitted from month 1 onward in participants with a serum potassium level of 4.8 mmol per liter or less and an eGFR that was no more than 30% less the value at the last visit. A decrease in the dose was allowed at any time for safety reasons. If the serum potassium level exceeded 5.5 mmol per liter, finerenone or placebo was withheld for 72 hours and restarted at a dose of 10 mg per day once the serum potassium level decreased to 5.0 mmol per liter or less. All the investigators, treating physicians, participants, and trial personnel were unaware of the trial-group assignments throughout.

OUTCOMES

The primary efficacy outcome was the total eGFR slope, defined as the mean annual rate of change in the eGFR from baseline to month 32. Exploratory eGFR outcomes included the mean annual rate of change in the eGFR from month 3 until discontinuation of the trial regimen (i.e., chronic eGFR slope) and the mean eGFR change from baseline to week 4 after discontinuation of finerenone or placebo. In exploratory analyses, the primary outcome was assessed in prespecified participant subgroups, which included subgroups defined according to stratification factors and baseline eGFR. Secondary outcomes, assessed according to a prespecified hierarchical testing strategy, were as follows: a composite of kidney or cardiovascular events (a sustained decrease of at least 57% in the eGFR [defined by a decrease

from baseline that was sustained for ≥ 4 weeks], kidney failure [defined by an eGFR of <15 ml per minute per 1.73 m^2 that was confirmed ≥ 4 weeks after the initial assessment, the need for long-term dialysis for ≥ 30 days, or kidney transplantation], hospitalization for heart failure, or death from cardiovascular causes), a composite of a sustained decrease from baseline of at least 57% in the eGFR or kidney failure as defined above, and a composite of hospitalization for heart failure or death from cardiovascular causes.

Additional exploratory outcomes included the change in the urinary albumin-to-creatinine ratio from baseline to month 6 and a composite of a sustained decrease of at least 40% in the eGFR (defined by a decrease from baseline that was sustained for ≥ 4 weeks) or kidney failure. Safety outcomes were adverse events and serious adverse events that began or worsened during the period between the first dose of finerenone or placebo and up to 3 days after any temporary or permanent interruption. Prespecified safety outcomes included a serum potassium level of more than 5.5 mmol per liter or more than 6.0 mmol per liter, hospitalization for serious hyperkalemia, and permanent discontinuation of the trial regimen because of hyperkalemia.

STATISTICAL ANALYSIS

We calculated that a sample size of 1500 participants would provide the trial with more than 90% power at a two-sided significance level of 5% to detect a mean annual rate of decline in the eGFR between baseline and month 32 that was 0.7 ml per minute per 1.73 m^2 less with finerenone than with placebo. This calculation assumed that 5% of the trial participants would have an intercurrent event (an event due to kidney failure or death from any cause) and that 5% would discontinue the trial.

Continuous data are expressed as means with standard deviations or medians with interquartile ranges, and categorical data are expressed as numbers and percentages. Efficacy analyses were conducted in the full analysis population, which included all the participants who underwent randomization. The primary efficacy outcome was analyzed with the use of a two-slope linear spline mixed-effects model that included trial group-specific fixed effects (finerenone vs. placebo) for the intercept, time from randomization,

time since month 3 (change point), and the randomization stratification factors and their interaction with the time components. The model included the fixed change point at month 3 to separate the effects of finerenone that occurred from baseline to month 3 (i.e., acute eGFR slope) from those that occurred from month 3 to discontinuation of the trial regimen, which were anticipated to be different. A two-sided P value of less than 0.05 was considered to indicate statistical significance.

Secondary outcomes were analyzed by means of stratified Cox proportional-hazards models and stratified log-rank tests fitted with the use of the randomization stratification factors. Cumulative incidence plots based on Aalen–Johansen estimates that accounted for competing risks (death from noncardiovascular causes or death from any cause, as appropriate) are presented according to trial group.

Data for participants with premature discontinuation or extended interruption of finerenone or placebo were handled with a treatment policy strategy. According to this strategy, participants continued to attend planned trial visits after discontinuation or interruption, and eGFR values collected during these visits were included in the primary efficacy analysis. Intercurrent events (dialysis, kidney transplantation, and death from any cause) were handled by means of a composite strategy, in which a multiple imputation approach was used to account for missing data after these events, and replacement data were obtained from a pool of unfavorable eGFR values. To assess the results of the primary efficacy analysis with respect to the effects of other missing data, a multiple imputation regression model that considered visit-specific trial-regimen discontinuation indicators and a tipping-point analysis that was based on multiple imputation were used.¹⁷ The primary outcome was also assessed with a shared parameter model. The shared-parameter model jointly models the eGFR slope and the times of intercurrent events in order to capture the association between these two outcomes (i.e., the risk of intercurrent events is higher among patients with a steeper eGFR slope). Details about the assumptions and methods used in the handling of missing data are provided in the Supplementary Appendix.

Safety outcomes were analyzed in the safety analysis population (defined as all the participants in the full analysis population who received at least one dose of finerenone or placebo). Analyses were performed with SAS software, version 9.4 or higher (SAS Institute).

RESULTS

PARTICIPANTS

Between September 21, 2021, and May 12, 2023, we assessed 3231 persons for eligibility. A total of 1584 participants underwent randomization — 793 were assigned to the finerenone group, and 791 to the placebo group. During a median treatment period of 36.6 months, 4 participants (0.5%) in the finerenone group and 2 (0.3%) in the placebo group discontinued the trial, and 157 (19.8%) and 177 (22.4%), respectively, discontinued the trial regimen (Fig. S1). Baseline characteristics and concomitant medications were balanced between the trial groups (Table 1 and Table S2). The mean age was 54.7 years; 535 participants (33.8%) were women, and the median urinary albumin-to-creatinine ratio was 818.9. Overall, 1580 participants (99.7%) were receiving a RAS inhibitor, and 270 (17.0%) were receiving an SGLT2 inhibitor. The representativeness of the trial population is shown in Table S3.

PRIMARY OUTCOME

At baseline, the mean ($\pm$ SD) eGFR was 46.8 ± 16.2 ml per minute per 1.73 m^2 in the finerenone group and 46.6 ± 16.0 ml per minute per 1.73 m^2 in the placebo group. Figure 1A shows the mean change in the eGFR from baseline over time in the trial groups. The mean rate of change in the eGFR from baseline to month 3 was 1.2 ml per minute per 1.73 m^2 greater with finerenone than with placebo (Fig. 1B). After month 3, the rate of eGFR decline was slower with finerenone.

At month 32, the mean eGFR was 39.4 ± 18.2 ml per minute per 1.73 m^2 with finerenone and 38.1 ± 18.4 ml per minute per 1.73 m^2 with placebo. In the analysis of the primary outcome, the mean annual rate of change in the eGFR from baseline to month 32 (total eGFR slope) was -3.3 ml per minute per 1.73 m^2 (95% CI, -3.6 to -3.1) with finerenone and -4.0 ml per minute per 1.73 m^2 (95% CI, -4.3 to -3.8) with placebo,

Table 1. Demographic and Clinical Characteristics of the Participants at Baseline.*

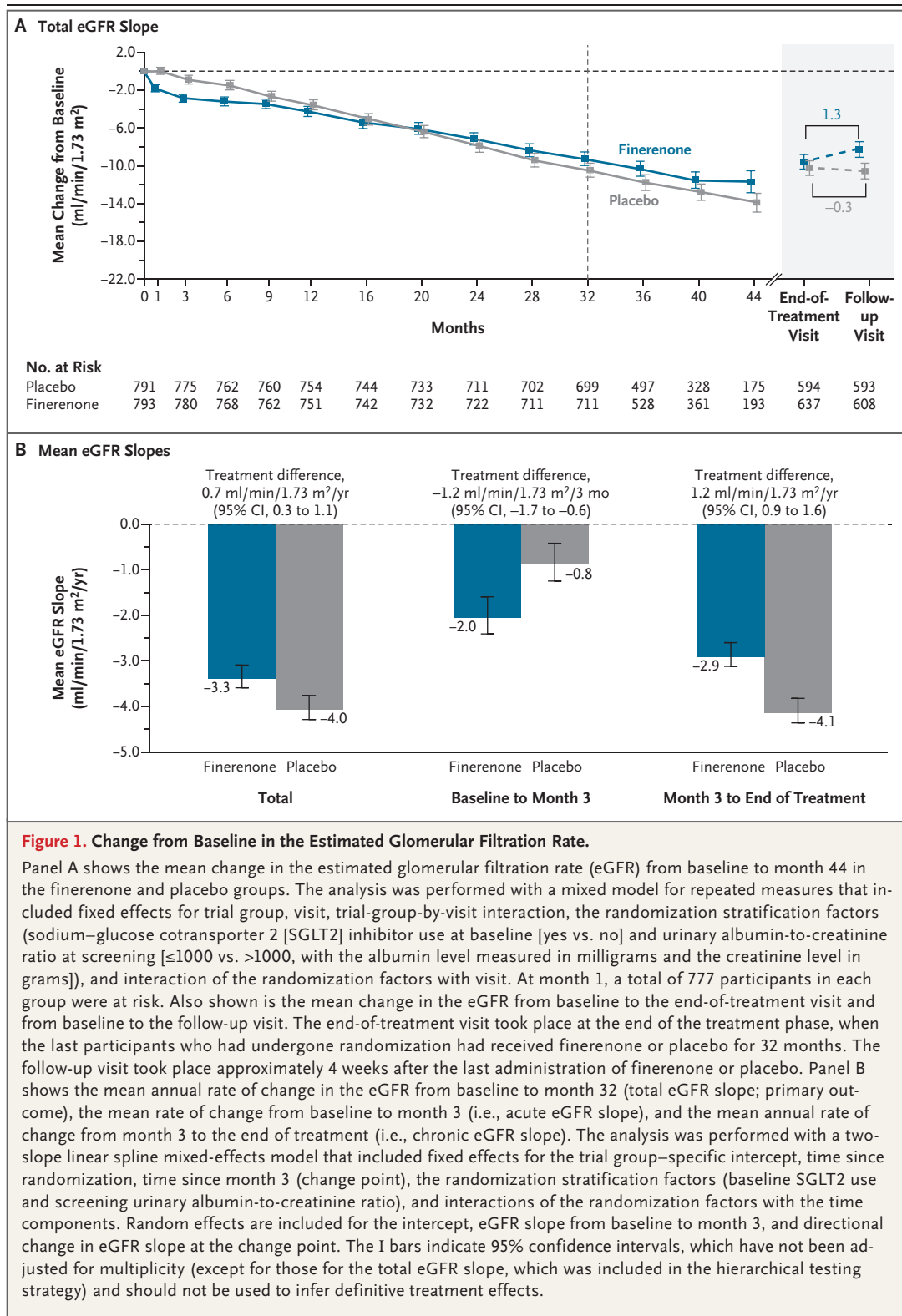
Characteristic	Finerenone (N=793)	Placebo (N=791)
Age — yr	54.5±14.2	54.8±14.3
Sex — no. (%)		
Female	275 (34.7)	260 (32.9)
Male	518 (65.3)	531 (67.1)
Race or ethnic group — no. (%)†		
Asian	447 (56.4)	421 (53.2)
Black	22 (2.8)	15 (1.9)
White	308 (38.8)	340 (43.0)
Other	16 (2.0)	15 (1.9)
Medical history		
Atherosclerotic disease	90 (11.3)	105 (13.3)
Hypertension	701 (88.4)	706 (89.3)
Heart failure	16 (2.0)	19 (2.4)
Primary cause of chronic kidney disease		
Hypertensive or ischemic nephropathy	234 (29.5)	225 (28.4)
Chronic glomerulonephritis	446 (56.2)	457 (57.8)
Other‡	29 (3.7)	24 (3.0)
Unknown	84 (10.6)	85 (10.7)
Clinical measurements — mm Hg		
Systolic blood pressure	129.7±13.9	129.3±14.2
Diastolic blood pressure	80.0±9.2	80.1±10.0
Biochemical measurements		
Urinary albumin-to-creatinine ratio§		
Median (IQR)	827.5 (572.0–1257.4)	810.7 (579.6–1234.3)
Distribution — no. (%)		
≤1000	483 (60.9)	509 (64.3)
>1000	310 (39.1)	282 (35.7)
Estimated glomerular filtration rate		
Mean — ml/minute/1.73 m ²	46.8±16.2	46.6±16.0
Distribution — no. (%)		
<60 ml/minute/1.73 m ²	631 (79.6)	636 (80.4)
≥60 ml/minute/1.73 m ²	162 (20.4)	155 (19.6)
Serum potassium level — mmol/liter	4.5±0.4	4.5±0.4
Medication use — no. (%)		
ACE inhibitor	230 (29.0)	205 (25.9)
Angiotensin-receptor blocker	560 (70.6)	585 (74.0)
Diuretic	142 (17.9)	148 (18.7)
SGLT2 inhibitor	135 (17.0)	135 (17.1)

* Plus-minus values are means ±SD. Percentages may not total 100 because of rounding. ACE denotes angiotensin-converting enzyme, IQR interquartile range, and SGLT2 sodium-glucose cotransporter 2.

† Race and ethnic group were reported by the participant. Other includes participants who reported American Indian or Alaska native race or native Hawaiian or other Pacific Islander race, did not report data on race or ethnic group (2 participants in the finerenone group and 4 participants in the placebo group), or reported more than one race or ethnic group.

‡ Other includes a primary cause of chronic kidney disease that was not classified as chronic glomerulonephritis and was not coded as “chronic kidney disease” or “nephropathy” with the *Medical Dictionary for Regulatory Activities* (MedDRA), version 28.1.

§ The albumin level was measured in milligrams, and the creatinine level was measured in grams.



corresponding to a decline that was 0.7 ml per minute per 1.73 m² (95% CI, 0.3 to 1.1) slower with finerenone ($P<0.001$) (Fig. 1B and Table 2). From the end of treatment to the end of follow-up, the mean eGFR increased with finerenone and decreased with placebo (Fig. 1A).

In the analysis of prespecified subgroups based on baseline characteristics, the effects of finerenone and placebo on the mean annual rate of decline in the eGFR were consistent with the effects shown in the primary efficacy analysis (Fig. 2 and Figs. S2 and S3). Evaluation of the total eGFR slope with the use of equations that included cystatin C level and a shared-parameter model showed results similar to those of the primary efficacy analysis (Fig. S4 and Table S4). The direction of the results was confirmed in prespecified sensitivity analyses that included multiple imputation with indicators of finerenone or placebo discontinuation and a tipping-point analysis (Tables S5 through S7).

SECONDARY OUTCOMES

Kidney or cardiovascular events (composite kidney–cardiovascular outcome) occurred in 110 participants (13.9%) in the finerenone group (4.7 events per 100 person-years; 95% CI, 3.9 to 5.6) and 134 participants (16.9%) in the placebo group (5.8 events per 100 person-years; 95% CI, 4.9 to 6.9), for a hazard ratio of 0.77 (95% CI, 0.60 to 0.99; $P=0.04$) (Table 2, Fig. 3, and Fig. S5). The hazard ratio (finerenone vs. placebo) for the composite of a sustained decrease from baseline of at least 57% in the eGFR or kidney failure was 0.78 (95% CI, 0.60 to 1.01; $P=0.06$), and the hazard ratio for the composite of hospitalization for heart failure or death from cardiovascular causes was 0.60 (95% CI, 0.27 to 1.33) (Table 2). The proportional-hazards assumption was assessed and was not violated.

EXPLORATORY EFFICACY OUTCOMES

The mean change in the eGFR from baseline to week 4 after the end of the treatment period was -8.8 ml per minute per 1.73 m² (95% CI, -9.7 to -7.8) with finerenone and -11.2 ml per minute per 1.73 m² (95% CI, -12.1 to -10.2) with placebo (difference, 2.4 ml per minute per 1.73 m²; 95% CI, 1.3 to 3.5). The relative change in the

urinary albumin-to-creatinine ratio from baseline to month 6 was -41.3% (95% CI, -44.2 to -38.3) with finerenone and -9.1% (95% CI, -13.1 to -5.0) with placebo (relative difference, 35.4%; 95% CI, 30.9 to 39.6); this apparent difference was sustained over time. A reduction of 30% or more in the urinary albumin-to-creatinine ratio from baseline to month 6 occurred in 444 participants (56.0%) in the finerenone group and 193 participants (24.4%) in the placebo group (odds ratio, 3.99; 95% CI, 3.22 to 4.95) (Fig. S6). The composite of a sustained decrease of at least 40% in the eGFR or kidney failure occurred in 168 participants (21.2%) in the finerenone group (7.4 events per 100 person-years; 95% CI, 6.3 to 8.5) and 206 participants (26.0%) in the placebo group (9.3 events per 100 person-years; 95% CI, 8.1 to 10.6), for a hazard ratio of 0.76 (95% CI, 0.62 to 0.93).

SAFETY OUTCOMES

Adverse events were reported in 542 participants (68.3%) who received finerenone and 517 participants (65.4%) who received placebo (Table 2). The percentage of participants with a serious adverse event was similar in the two groups (20.9% [166 participants] with finerenone and 21.2% [168 participants] with placebo). Adverse events leading to permanent discontinuation of the trial regimen occurred in 17 participants (2.1%) in the finerenone group and 23 participants (2.9%) in the placebo group; serious adverse events leading to death occurred in 6 (0.8%) and 10 (1.3%), respectively. Adverse events according to *Medical Dictionary for Regulatory Activities*, version 28.1, primary system organ class are shown in Table S8. Death from any cause occurred in 19 participants (2.4%) in the finerenone group and 28 participants (3.5%) in the placebo group.

Hyperkalemia was reported in 135 participants (17.0%) in the finerenone group and 105 participants (13.3%) in the placebo group. Overall, the incidence of any serious hyperkalemia event was less than 1%. Hyperkalemia led to hospitalization in 7 participants (0.9%) with finerenone and 5 participants (0.6%) with placebo and to permanent discontinuation of the trial regimen in 12 (1.5%) and 1 (0.1%), respectively. No fatal hyperkalemia events occurred. A greater increase in the

Table 2. Outcomes and Adverse Events.

Variable	Finerenone (N = 793)	Placebo (N = 791)	Treatment Effect*
Primary outcome			
Mean total eGFR slope (95% CI)	-3.3 (-3.6 to -3.1)	-4.0 (-4.3 to -3.8)	0.7 (0.3 to 1.1)†
Secondary outcomes — no. (%)			
Composite kidney–cardiovascular outcome‡	110 (13.9)	134 (16.9)	0.77 (0.60 to 0.99)§
Composite kidney outcome¶	104 (13.1)	125 (15.8)	0.78 (0.60 to 1.01)
Composite cardiovascular outcome**	10 (1.3)	16 (2.0)	0.60 (0.27 to 1.33)
Adverse events — no. (%)			
Any adverse event			
Overall	542 (68.3)	517 (65.4)	
Related to trial regimen	175 (22.1)	133 (16.8)	
Leading to permanent discontinuation of the trial regimen	17 (2.1)	23 (2.9)	
Any serious adverse event			
Overall	166 (20.9)	168 (21.2)	
Related to trial regimen	9 (1.1)	6 (0.8)	
Leading to permanent discontinuation of trial regimen	1 (0.1)	8 (1.0)	
Leading to hospitalization	159 (20.1)	156 (19.7)	
Life-threatening	4 (0.5)	8 (1.0)	
Leading to death	6 (0.8)	10 (1.3)	
Acute kidney injury			
Overall	23 (2.9)	25 (3.2)	
Leading to permanent discontinuation of the trial regimen	1 (0.1)	1 (0.1)	
Serious adverse event leading to hospitalization	9 (1.1)	8 (1.0)	
Any hyperkalemia††			
Overall	135 (17.0)	105 (13.3)	
Leading to permanent discontinuation of the trial regimen	12 (1.5)	1 (0.1)	
Serious adverse event related to hyperkalemia			
Overall	8 (1.0)	5 (0.6)	
Leading to hospitalization	7 (0.9)	5 (0.6)	
Life-threatening	1 (0.1)‡‡	0	
Leading to death	0	0	

* The treatment effect for the primary outcome is the between-group difference in the total eGFR slope (mean annual rate of change in the eGFR from baseline to month 32). The treatment effect for the secondary outcomes is the hazard ratio (finerenone vs. placebo) for the indicated outcome event. Treatment effects are presented with 95% confidence intervals.

† P<0.001 for the difference in the mean total eGFR slope.

‡ The composite kidney–cardiovascular outcome was a sustained decrease of at least 57% in the eGFR (defined by a decrease from baseline that is sustained for ≥4 weeks), kidney failure (defined by an eGFR of <15 ml per minute per 1.73 m² of body-surface area that was confirmed ≥4 weeks after the initial assessment, the need for long-term dialysis for ≥30 days, or kidney transplantation), hospitalization for heart failure, or death from cardiovascular causes.

§ P=0.04 for the difference in the risk of the composite of kidney or cardiovascular events.

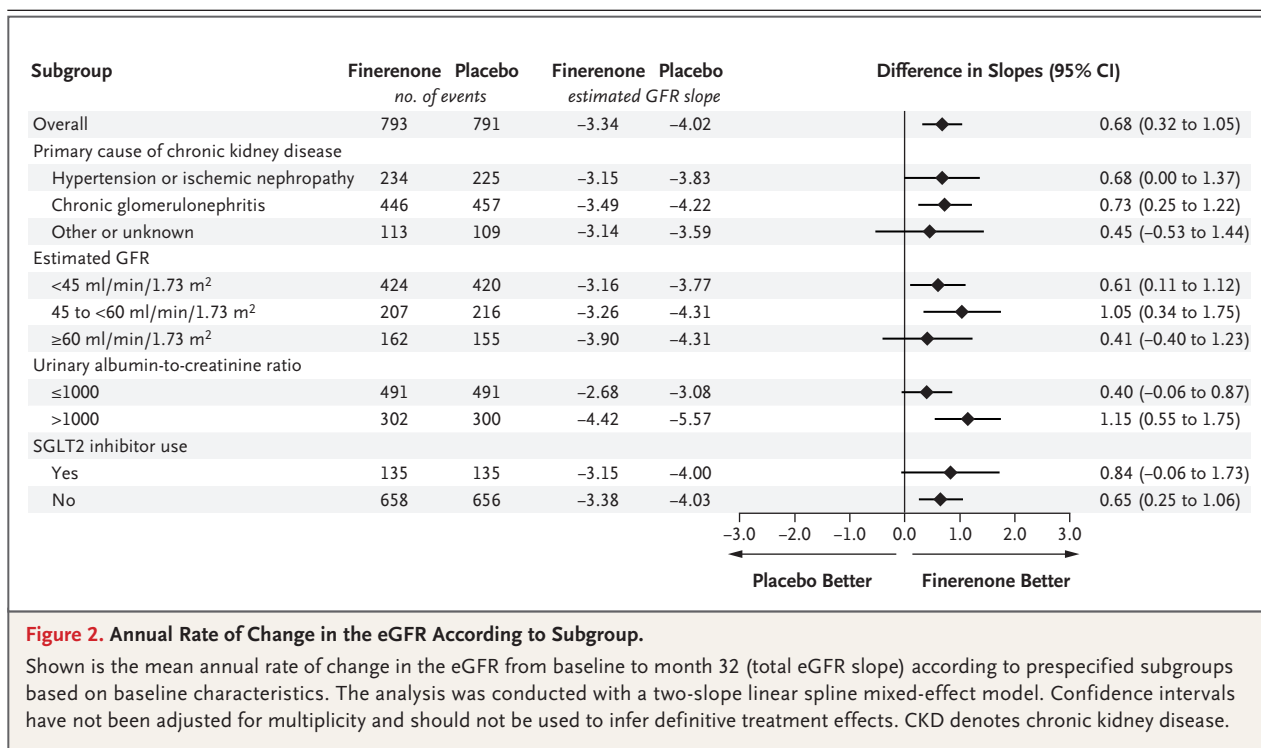
¶ The composite kidney outcome was a sustained decrease of at least 57% in the eGFR or kidney failure as defined above.

|| P=0.06 for the difference in the risk of the composite of kidney events.

** The composite cardiovascular outcome was hospitalization for heart failure or death from cardiovascular causes.

†† Hyperkalemia was an adverse event of special interest and defined as any adverse event reported by the investigator and coded with the MedDRA preferred terms of hyperkalemia or blood potassium increased. Any hyperkalemia related to the trial regimen occurred in 94 participants (11.9%) in the finerenone group and 74 participants (9.4%) in the placebo group. Serum potassium levels were measured at a central laboratory. A serum potassium level of more than 5.5 mmol per liter was present in 146 of 776 participants (18.8%) in the finerenone group and 95 of 776 participants (12.2%) in the placebo group; values were missing for 1 and 7 participants, respectively. A serum potassium level of more than 6.0 mmol per liter was present in 34 of 786 participants (4.3%) in the finerenone group and 18 of 784 participants (2.3%) in the placebo group; values were missing for 5 participants in the placebo group.

‡‡ The participant had a serum potassium level of 7 mmol per liter and was not hospitalized.



mean serum potassium level from baseline to month 1 was observed with finerenone than with placebo (difference, 0.12 mmol per liter); potassium levels in the finerenone group stabilized thereafter (Fig. S7). Serum potassium levels of more than 5.5 mmol per liter were reported in 146 of 776 participants (18.8%) in the finerenone group and 95 of 776 participants (12.2%) in the placebo group; levels of more than 6.0 mmol per liter occurred in 34 of 786 (4.3%) and 18 of 784 (2.3%), respectively. The incidence of acute kidney injury was similar in the trial groups; few of these events (<0.5%) led to permanent discontinuation of finerenone or placebo.

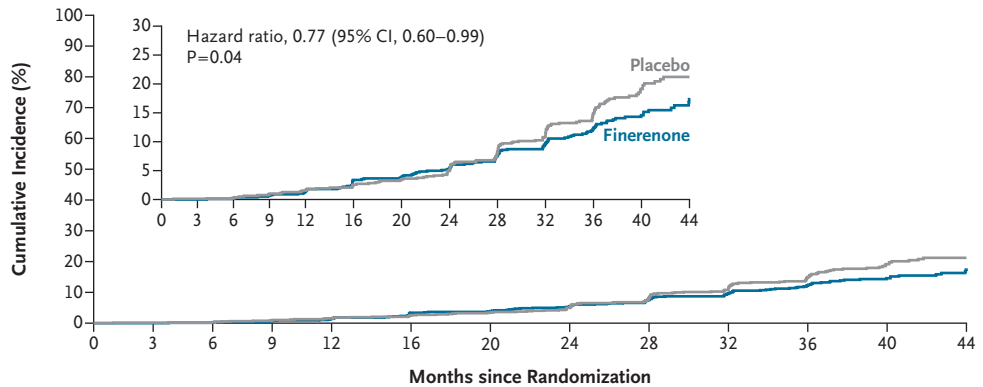
The mean change in systolic blood pressure from baseline to month 3 was -5.1 mm Hg with finerenone and -0.1 mm Hg with placebo (Fig. S8). The mean change in diastolic blood pressure from baseline to month 3 was -3.1 mm Hg and less than -0.1 mm Hg, respectively (Fig. S9).

DISCUSSION

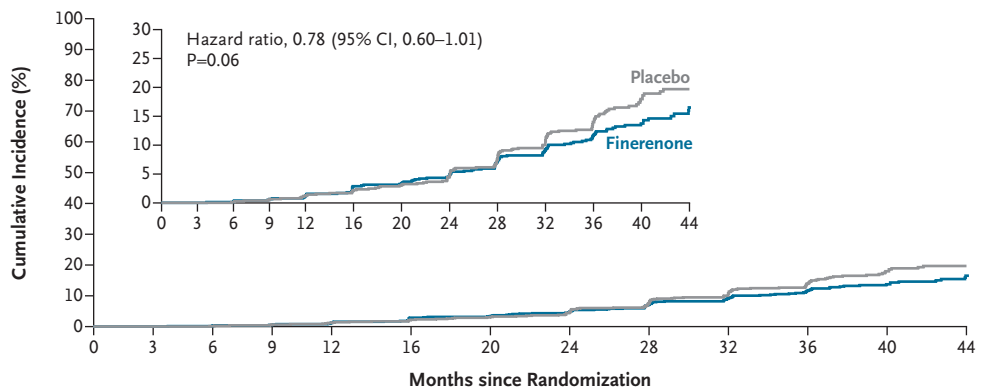
In this double-blind, randomized, placebo-controlled clinical trial, finerenone resulted in

a slower annual rate of eGFR decline in participants without diabetes who had CKD and were receiving an angiotensin-converting enzyme inhibitor or angiotensin-receptor blocker.¹⁰ The incidence of the composite of kidney or cardiovascular events was lower with finerenone than with placebo. Hyperkalemia events were more frequent with finerenone than with placebo, but serious hyperkalemia events were infrequent.

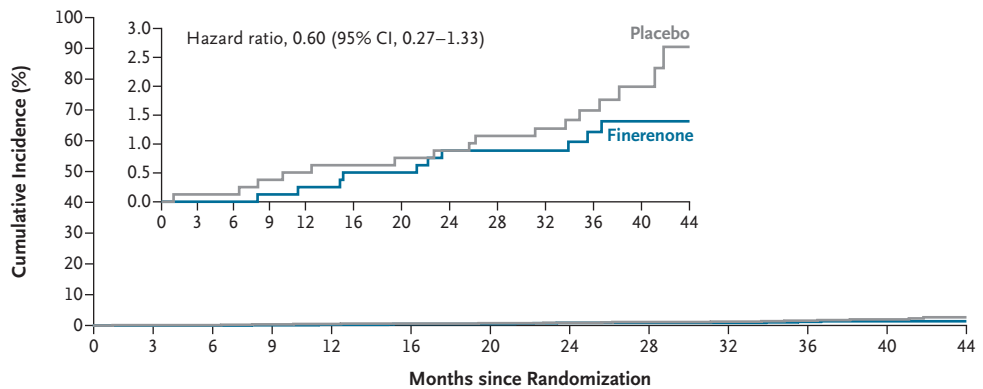
The effect of finerenone on preserving kidney function was previously reported in persons with CKD and type 2 diabetes.¹³⁻¹⁵ In the FIDELIO-DKD (Finerenone in Reducing Kidney Failure and Disease Progression in Diabetic Kidney Disease) trial, the decline in the mean annual rate of change in the eGFR was 0.7 ml per minute per 1.73 m² slower with finerenone than with placebo.¹³ In our trial, finerenone treatment resulted in similar attenuation of the rate of eGFR decline as compared with placebo. The magnitude of the effect of finerenone in the current trial was similar to the magnitude previously reported with other kidney-protective drugs, including RAS inhibitors and SGLT2 inhibitors, which reduced the annual rate of decline in the eGFR by 0.5 to 1.0 ml per minute per 1.73 m².^{5,18-20} The reduction in the

A Composite Kidney–Cardiovascular Outcome**No. at Risk**

Placebo	791	782	778	766	755	744	731	714	680	624	429	253	124
Finerenone	793	786	777	767	753	735	724	708	688	639	471	305	152

B Composite Kidney Outcome**No. at Risk**

Placebo	791	783	779	768	757	746	733	717	683	628	433	255	125
Finerenone	793	786	777	768	755	739	728	714	692	643	473	306	152

C Composite Cardiovascular Outcome**No. at Risk**

Placebo	791	790	789	785	781	778	775	771	767	735	545	334	166
Finerenone	793	792	791	789	786	783	780	776	775	744	554	358	180

Figure 3 (facing page). Key Secondary Efficacy Outcomes.

Shown are cumulative incidence plots for a composite of kidney or cardiovascular events (a sustained decrease of at least 57% in the eGFR [defined by a decrease from baseline that is sustained for ≥ 4 weeks], kidney failure [defined by an eGFR of <15 ml per minute per 1.73 m² of body-surface area that was confirmed ≥ 4 weeks after the initial assessment], the need for long-term dialysis for ≥ 30 days, or kidney transplantation), hospitalization for heart failure, or death from cardiovascular causes (composite kidney–cardiovascular outcome) (Panel A). Also shown are cumulative incidence plots for a composite of a sustained decrease of at least 57% in the eGFR or kidney failure as defined above (composite kidney outcome) (Panel B) and a composite of hospitalization for heart failure or death from cardiovascular causes (composite cardiovascular outcome) (Panel C). Cumulative incidence plots were based on Aalen–Johansen estimates that accounted for death from noncardiovascular causes or death from any cause as a competing event. Insets show the same data on an expanded y axis. Confidence intervals have not been adjusted for multiplicity and should not be used to infer definitive treatment effects.

mean annual rate of change in the eGFR from baseline to month 3 with finerenone suggests that the total eGFR slope may be an underestimate because finerenone, by means of its mechanism of action, initially decreases the eGFR. This early change is temporary, and although it may have affected the between-group difference in the total eGFR slope, it did not affect the difference in the mean change in the eGFR from month 3 until discontinuation of the trial regimen (1.2 ml per minute per 1.73 m²).

The primary outcome was the change in the eGFR over time, a surrogate end point that is increasingly being used to assess drug efficacy in CKD trials.²¹ Although our trial was not powered to assess the effects of the trial regimens on clinical outcomes, it showed that, in addition to the observed reduction in the annual rate of eGFR decline with finerenone, the risk of the composite of kidney or cardiovascular events was lower in the finerenone group than in the placebo group. This finding is similar to the results of a meta-analysis of 66 randomized controlled trials, which found that an attenuation of 0.75 ml per minute per 1.73 m² in the eGFR decline as compared with placebo predicted a reduction of 23% in the risk of adverse kidney outcomes.²¹

During trial recruitment, updated clinical practice guidelines recommended the use of SGLT2

inhibitors as part of evidence-based therapy in patients with CKD.^{10,22} In our trial, 17% of the participants were using SGLT2 inhibitors at baseline,¹⁶ and the efficacy of finerenone in participants who were using SGLT2 inhibitors was similar to that in participants who were not using SGLT2 inhibitors. Although we hypothesize that the use of both finerenone and SGLT2 inhibitors may provide complementary effects, this was not directly tested in the present trial.

Use of the steroidal mineralocorticoid receptor antagonists spironolactone and eplerenone in persons with CKD can be limited by the side-effect profile of these drugs and safety concerns, particularly hyperkalemia.²³ Of the approximately 1400 participants with CKD in the BARACK-D (Benefits of Aldosterone Receptor Antagonism in Chronic Kidney Disease) trial, more than half discontinued spironolactone within 6 months after randomization owing to safety criteria (primarily related to acute kidney injury) that were prespecified in the protocol; hyperkalemia accounted for more than 10% of the discontinuations.²⁴ No unanticipated safety signals were observed with finerenone in the current trial. Hyperkalemia was the most common adverse event, but increases in the serum potassium level were modest, leading to discontinuation of the trial regimen in few participants and no hospitalizations. The incidence rates of acute kidney injury and symptomatic hypotension were also low and similar in the finerenone and placebo groups.

This trial has certain limitations. Although we recruited a geographically diverse population, most of the participants were men and had advanced CKD with severe albuminuria, and few were Black. Furthermore, participants with kidney diseases such as autosomal dominant or autosomal recessive polycystic kidney disease, lupus nephritis, or antineutrophil cytoplasmic antibody-associated vasculitis were excluded, as were those with an extremely low eGFR or low-level albuminuria. Thus, the findings here are applicable to persons with characteristics similar to those of the trial participants and cannot be generalized.

In the current trial involving adults without diabetes who had CKD, finerenone resulted in a slower rate of eGFR decline over 32 months than placebo.

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A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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